A. Venkata Sai Tejesh

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TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, Reactjsx
Backend: Node.js, Fastapi, Express.js
Database: MongoDB,MySQL **Version Control:** Git, GitHub
Programming Languages: C++
Libraries: PyTorch, NLTK, scikit-learn pandas, NumPy, OpenCV, Librosa,Langchain etc..

AI/ML Projects

. Agentic system

→ Built an **intelligent multi-agent AI system** that classifies queries into general Q&A, document-based (RAG), or tool-invocation tasks.

→ Enabled **document querying with RAG**, allowing users to upload documents and get precise, context aware answers.

→ Implemented **Redis-based session management** to preserve conversation history and maintain context across multiple turns.

→ Applied **memory optimization techniques** (relevant chunk extraction, hallucination mitigation) to handle long conversations efficiently.

→ Structured the system using a **modular multi-agent architecture** (classification, orchestration, and response refinement), ensuring scalability and easy extensibility.

· **Text Summarizer Using LSTM**

→ Implemented a sequence-to-sequence model with LSTM units in both encoder and decoder. → Used pretrained GloVe embeddings to convert words to embeddings and mapped them to indices. → Utilized an embedding matrix to feed the embeddings into the LSTM layers. → Utilised a subset of the CNN/DailyMail dataset for abstractive summarization.

· **Sound Classification using CNN (PyTorch)**

→ Applied preprocessing: amplitude normalization, noise reduction, segmentation, and resampling to 16kHz.

→ Performed DFT(Discrete Fourier transform) and windowing to generate raw spectrogram data. → Spatial features learned by CNN are fed to the Fully Connected (NN) for classification → Converted spectrogram data into 2D images for CNN input and achieved 94% accuracy .

Web Development Projects

· **Doctor-Patient Interaction Website**

→ Developed a real-time medical portal for patients and doctors using WebSockets. → Enabled secure chat with image, video, and document sharing in private rooms. → Designed and implemented backend with Node.js, Express.js, and MongoDB. → Built appointment system and pharmacy section showing drug details.

→ Integrated JWT for secure session authentication and role-based access control. ·

Content-Management-System

→ Built a multi-role platform with Admin, Viewer, and Editor workflows.

→ Integrated advanced editor (Tiptap) with image upload and rich text features.

→ Implemented real-time post submission & approval using WebSockets.

→ Designed notification system with status tracking (approved, pending, drafts). → Enabled editors to approve/reject posts with comments and viewers to revise drafts.

WORK EXPERIENCE:

Intern – College R&D Department -2months

- Developed and maintained the college R&D website [RndSite].
- Technologies used: [Reactjsx,tailwindcss,strapi(cms),nodejs etc..]

PORTFOLIO:

GitHub: github.com/bunny-learner

EDUCATION

B.Tech, Computer Science & Engineering 2023 – 2027 Indian Institute of Technology Dharwad CGPA: 7.93/10

Senior Secondary (Inter Second Year), SSE (State Board) 2023 Narayana Residential Junior College, Brahmadevam Campus, Nellore, AP Percentage: 96.00%

Secondary (X), SSE (State Board) 2020 Nagarjuna Model School, Kadapa, AP Percentage: 94.33%

ADDITIONAL DETAILS

· Secured AIR 4882 among 180,372 candidates in JEE Advanced 2023

· Secured AIR 8322 among 1,113,325 candidates in JEE Mains 2023